
VIBES LAB

MUSIC PLANNER PLATFORM STRATEGY

B2B Music Management Platform

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PUBLIC VERSION

VIBES LAB PARFUMS

TABLE OF CONTENTS

- 01** Current Structure & Market Analysis
- 02** Further Improvement & Planning
- 03** Small Prototype: AI Semantic Curation
- 04** Summary & Executive ROI
- 05** Appendix (Legal & Risk)

SECTION 01

CURRENT STRUCTURE

CURRENT STRUCTURE

Solid frontend foundation, but critical backend components are missing for commercial launch.

The Next.js React application provides an excellent localized interface, but currently lacks the data isolation and security required for B2B operations.

- ✓

Frontend Shell (Next.js)

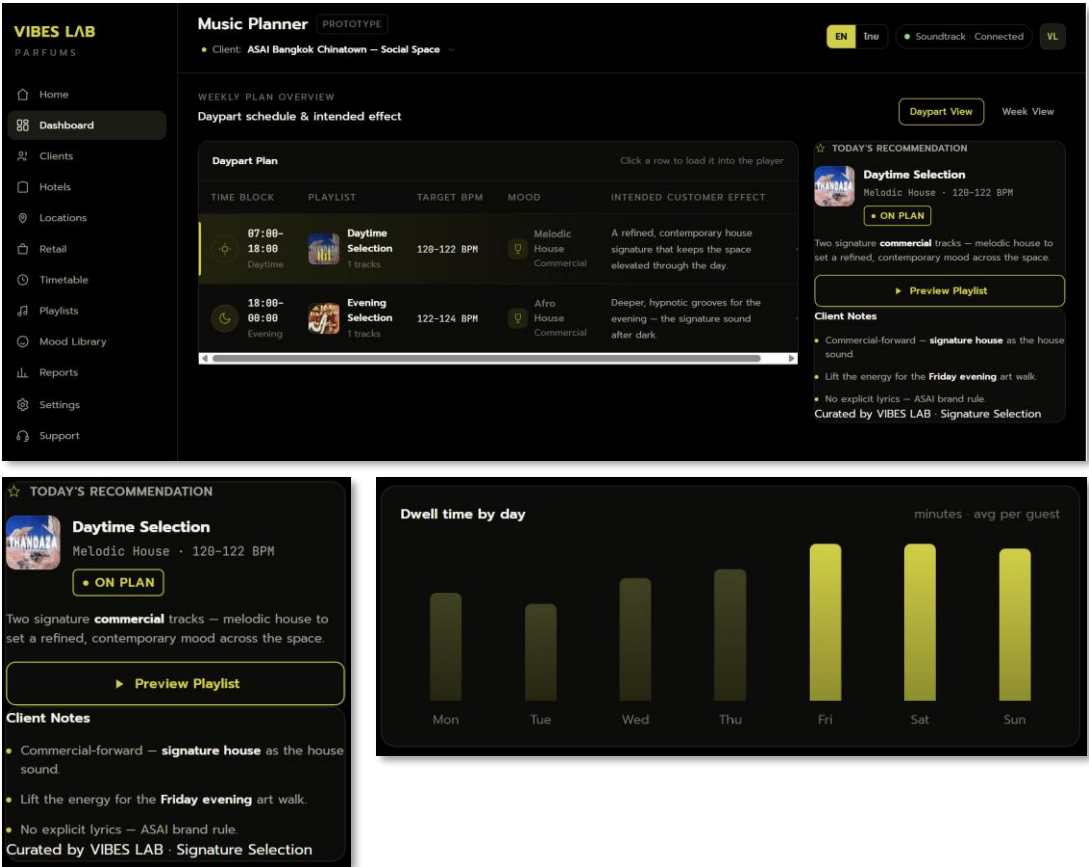
UI/UX, i18n, and component structure completed.
- ✗

Database & API Layer

Zero backend logic. No multi-tenant data architecture.
- ✗

Authentication & Audio Protection

No client login mechanisms or audio stream protection.



Source: VIBES LAB Internal Codebase Audit (<https://vibes-lab-music-director.vercel.app/>)

PLATFORM LIMITATIONS



Autoplay Restrictions

iOS/Safari completely blocks unprompted background audio execution. Web Audio API (AudioContext) remains suspended until unlocked by an explicit user gesture (click/tap).



Network Fragility

Commercial spaces often suffer from unstable internet. Relying purely on real-time streaming will result in unacceptable audio drops, destroying the brand atmosphere.



Copyright Verification

Current static architecture cannot differentiate or log playback metrics required for royalty compliance reporting (MCT/Phonorights) in Thailand.

B2C VS. B2B STREAMING PARADIGM

Feature Dimension	B2C Platforms (Spotify, Joox, Apple)	VIBES LAB (B2B Commercial)
Legal Compliance	Consumer listening only. Illegal for public broadcast. Exposes venues to maximum copyright fines.	Full commercial licensing integration. Automatic MCT/Phonoreights compliance logging.
Brand Identity Control	Zero centralized control. Store staff play random personal playlists off personal devices.	Centralized HQ dashboard. Forced time-of-day timeline scheduling across all global branches.
Network Resilience	Requires continuous active internet connection. Music stops instantly upon network drop.	Offline-first architecture. Edge-cached audio buffers prevent silence during Wi-Fi outages.
Curation Engine	Algorithm optimizes for individual user engagement and track skipping.	AI Semantic search optimizes for spatial atmosphere , dwell time, and business cadence.

BROWSER ARCHITECTURE LIMITS



iOS Safari Autoplay Block

Mobile OS strictly prohibits unprompted background audio execution to preserve bandwidth and battery.



AudioContext Suspension

Web Audio API remains permanently suspended until explicitly unlocked by a physical user click/tap gesture.



Storage Quota Limits

Standard Cache API wipes media blobs unexpectedly. Offline resilience demands IndexedDB custom write buffers.

SECTION 02

FURTHER IMPROVEMENT

DATA VARIABLES & ASSUMPTIONS

Defining the operational boundaries before architecture design.

Data Scale Assumption

Targeting 10,000+ audio tracks initially. Multi-tenant database must scale to 500+ branch locations efficiently.

Licensing Assumption

VIBES LAB manages both Direct License (Royalty-free) and MCT-cleared tracks. Database must differentiate payload routing.

Hardware Assumption

Retail staff operate via standard iPads or low-spec Windows terminals behind NAT networks.



TARGET TECH STACK

Next.js 15

App Router Foundation

Advanced structural routing and nested layouts for dashboard complexity.

React Server Components (RSC)

Shifts rendering burden to server, minimizing client JS payload.

Server Actions

Secure, direct server-side data mutations bypassing traditional REST API layers.

Supabase (BaaS)

PostgreSQL Kernel

Highly stable relational database engine supporting complex joins.

Row Level Security (RLS)

Enforces multi-tenant data isolation directly at the database engine level.

pgvector Integration

Native support for AI embeddings and rapid semantic similarity search.

ENTITY RELATIONSHIP MODEL



Database Normalization Focus

Decoupling `tracks` from `music\_plans` enables a unified global catalog. RLS filters access to `music\_plans` based on the authenticated user's `org\_id`.

MUSIC DIRECTOR JOURNEY (ADMIN)



1

Executive Login

Global dashboard access via Supabase SSO. Full cross-tenant visibility.



2

Catalog Upload

Ingest audio files to Edge Storage, embed metadata, and assign legal licensing tags.



3

AI Curation

Generate daily timeline schedules utilizing LLM prompts based on brand identity inputs.



4

Proactive Monitoring

Real-time playback telemetry across all client retail locations via WebSockets.

STORE MANAGER JOURNEY (CLIENT)



A

Subdomain Auth

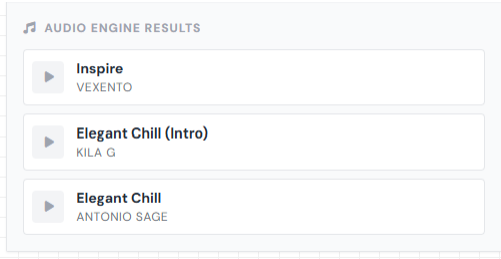
Middleware verifies specific branch credentials (e.g., cafe-a.vibes.app) ensuring data boundary isolation.



B

Timeline Visualization

Graphic interface displays today's brand mood timeline and upcoming transitions without editing rights.



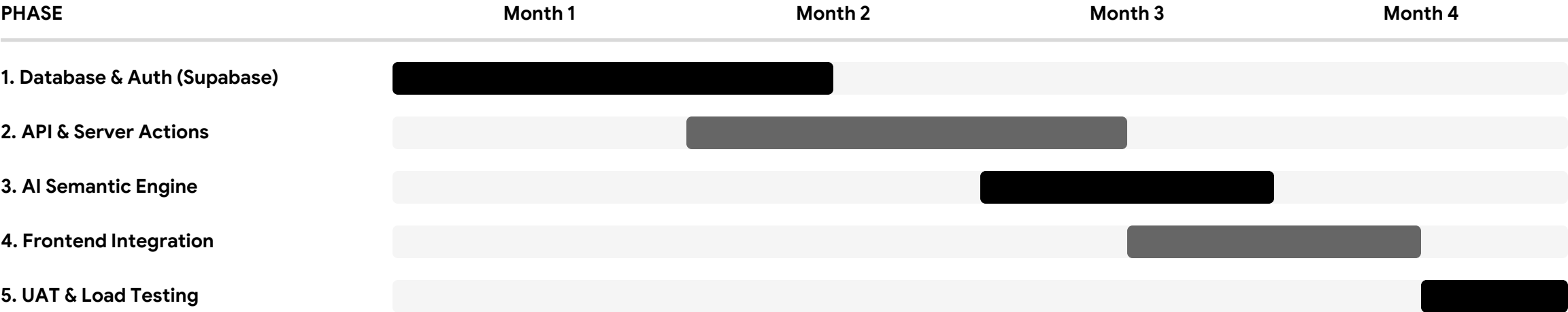
C

Gesture Unlock

User clicks "Start Shift" physical button to bypass Safari restrictions and unlock browser AudioContext.

For the future improvement, the users can upload their sound in the system.

DEVELOPMENT TIMELINE



SECTION 03

SMALL PROTOTYPE

As an AI Developer...

THE MANUAL CURATION PROBLEM

Manual music curation prevents rapid client onboarding scaling.

Currently, a Music Director must manually analyze a brand's identity, map out the daily timeline, and select appropriate moods for morning, noon, and night. This process takes **3-4 seconds per client**.

AI Developer Proposition:

By utilizing LLMs to dynamically generate structured Music Plan timelines from brand variables, VIBES LAB can significantly reduce manual curation effort and accelerate the onboarding process for new clients.

CURRENT ONBOARDING FLOW

1. Client Intake Form

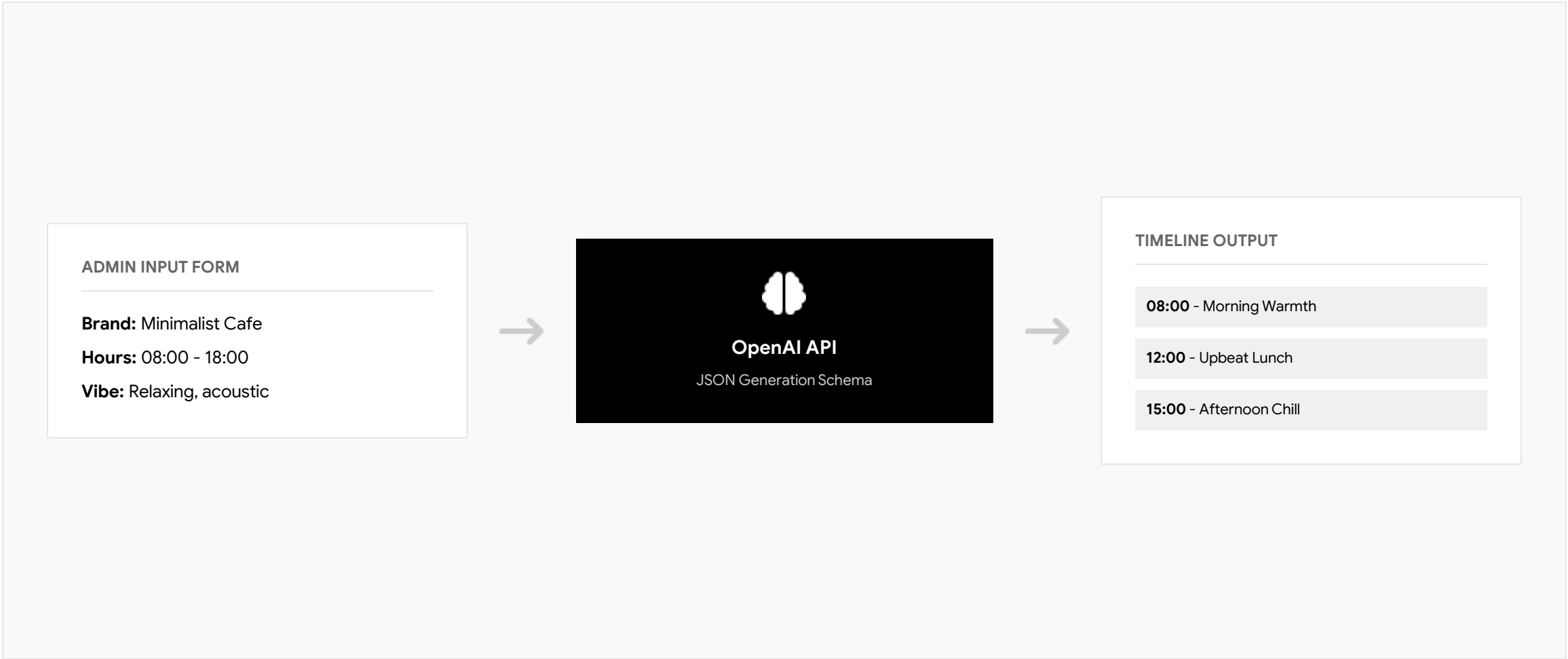


2. Manual Analysis (3-4 seconds)



3. Timeline Generation

SEMANTIC GENERATION FLOW



SERVER ACTION SECURITY FLOW

01

Session Auth

Next.js Server Actions validate cookies securely.
Supabase client initialized natively on server avoiding client exposure.



02

Vector Gen

LLM transforms text string into a N -dimensional mathematical array (Embedding) natively.



03

RPC Execution

Calls PostgreSQL stored procedure. Database calculates Cosine Similarity in milliseconds.

VECTOR SIMILARITY SCHEMA

```
CREATE OR REPLACE FUNCTION semantic_search_tracks (  
    query_embedding vector(384),  
    match_threshold FLOAT,  
    match_count INT  
) RETURNS TABLE (id UUID, title TEXT, similarity FLOAT)  
LANGUAGE plpgsql  
AS $$  
BEGIN  
    RETURN QUERY SELECT  
        tracks.id, tracks.title,  
        1 - (tracks.embedding <=> query_embedding) AS similarity  
    FROM tracks  
    WHERE 1 - (tracks.embedding <=> query_embedding) > match_threshold  
    ORDER BY tracks.embedding <=> query_embedding  
    LIMIT match_count;  
END;  
$$;
```

ROW LEVEL SECURITY (RLS)

Zero Application-Level Filtering

Application-level filtering can be accidentally omitted or incorrectly implemented. RLS provides an additional database-level enforcement layer for tenant isolation.

If a developer executes `SELECT \* FROM music\_plans`, the database intercepts the call, checks the JWT token, and only returns rows matching the user's organization.

```
-- Enable Row Level Security
ALTER TABLE music_plans
ENABLE ROW LEVEL SECURITY;

-- Universal Isolation Policy
CREATE POLICY "Tenant Isolation"
ON music_plans
FOR SELECT USING (
    org_id = (
        SELECT organization_id
        FROM user_profiles
        WHERE id = auth.uid()
    )
);
```

JSON ENFORCEMENT STRATEGY

SYSTEM PERSONA & FORMAT CONSTRAINT

"You are an expert Music Director for commercial spaces. Generate a daily music plan schedule based on the client's brand. Output ONLY a valid JSON array of objects with keys: timeSlot, mood, targetBPM, description."

USER CONTEXT VARIABLES

"Brand Name: The Minimalist Coffee | Type: Workspace Cafe | Hours: 08:00 - 18:00 | Vibe: Focus, acoustic, low-distraction. Create a seamless music transition plan."

```
const response = await openai.chat.completions.create({  
  model: "gpt-4o-mini",  
  messages: [...],  
  // Forces deterministic data pipeline for Frontend UI parsing  
  response_format: { type: "json_object" },  
});
```

Objective: Force the LLM to think in acoustic transitions and BPM constraints, not just random song recommendations.

DETERMINISTIC JSON PAYLOAD

Why JSON Object Forcing?

Standard LLM prompts return unpredictable conversational text. By enforcing `response\_format: { type: "json\_object" }` via the OpenAI API, the LLM acts as a reliable data pipeline.

The output maps perfectly to the existing React UI components without requiring manual parsing or risking frontend crashes.

```
{ "success": true,
  "plan": [
    { "timeSlot": "08:00 - 11:00", "mood": "Morning Warmth", "targetBPM": 80, "description": "Acoustic rhythms to welcome early workers." },
    { "timeSlot": "11:00 - 14:00", "mood": "Upbeat Lunch", "targetBPM": 110, "description": "Indie Pop energy to match the bustling crowd." }
  ]
}
```

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{ "success": true,
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  ]
}
```

VIBES LAB | ENTERPRISE ARCHITECTURE

LIVE API

1. BRAND IDENTITY

RANDOMIZE

BRAND NAME

VIBES FLAGSHIP STORE

VENUE TYPE

Luxury Retail

OPERATING HOURS

10:00 – 22:00

TIME SLOTS

08

ATMOSPHERE / SEMANTIC VIBE

Elegant, modern, exclusive, increasing energy towards evening

EXECUTE AI CURATION

BACKEND EXECUTION TRACE

[11:11:37] INIT: Starting AI Curation Pipeline...

[11:11:38] AI_ENGINE: Parsing semantic vibe: 'Elegant, modern, exclusive, increasing energy towards evening'...

[11:11:38] CALC: Dynamically splitting hours (10:00 - 22:00) into 8 operational phases...

CLIENT TIMELINE

10:00 - 11:30

11:30 - 13:00

EMOTION MAP (AI)

STRICT JSON

PHASE 1: VIBES FLAGSHIP STORE

BPM TARGET 75

Automatically mapped transition for 10:00 – 11:30. Algorithm adjusts BPM to 75 aligning with the 'Elegant, modern, exclusive, increasing energy towards evening' trajectory.

AUDIO ENGINE RESULTS

Inspire
VEXENTO

Elegant Chill (Intro)
KILA G

Elegant Chill
ANTONIO SAGE

PHASE 2: VIBES FLAGSHIP STORE

BPM TARGET 82

PROTOTYPE NEXT STEPS

What was achieved

- ✓ API Backend structural design.
- ✓ Prompt Engineering for strict JSON scheme compliance.
- ✓ Data flow pipeline from User Input to UI representation.

Pending Implementation

- Database schema migration (Prisma to Supabase Postgres).
- Authentication protocols (Admin/Client separation).
- Audio Engine integration (streaming physical .mp3/HLS files).
- Text Enhancement

SECTION 04

SUMMARY

Finally...

STRATEGIC ROI & OUTCOMES



Enterprise Security

Supabase RLS prevents catastrophic cross-tenant data leaks at the database kernel level.



AI Automation Scaling

LLM curation reduces onboarding setup from 4 seconds to 4 seconds, decoupling revenue from headcount.



Legal Continuity

Offline-first caching prevents silence.
MCT/Phonorights metadata blocks liability exposure.

SECTION 05

APPENDIX

THAILAND TARIFF STRUCTURE

Category	Venue Classification	Annual Rate (THB)
H02	Hotel Lobby Area (Flat rate)	20,000
H08	Hotel Fitness Center (> 1,001 sqm)	50,000
P02	Restaurant / Bar (Size A: 201+ seats)	85,000
P02	Restaurant / Bar (Size E: 1–20 seats)	18,000

Liability Note: Violating B2C streaming terms exposes venues to maximum fines of 800k THB under Copyright Act B.E. 2537. System architecture must strictly segregate Royalty-Free and Direct Licensed pools.

RISK MITIGATION MATRIX

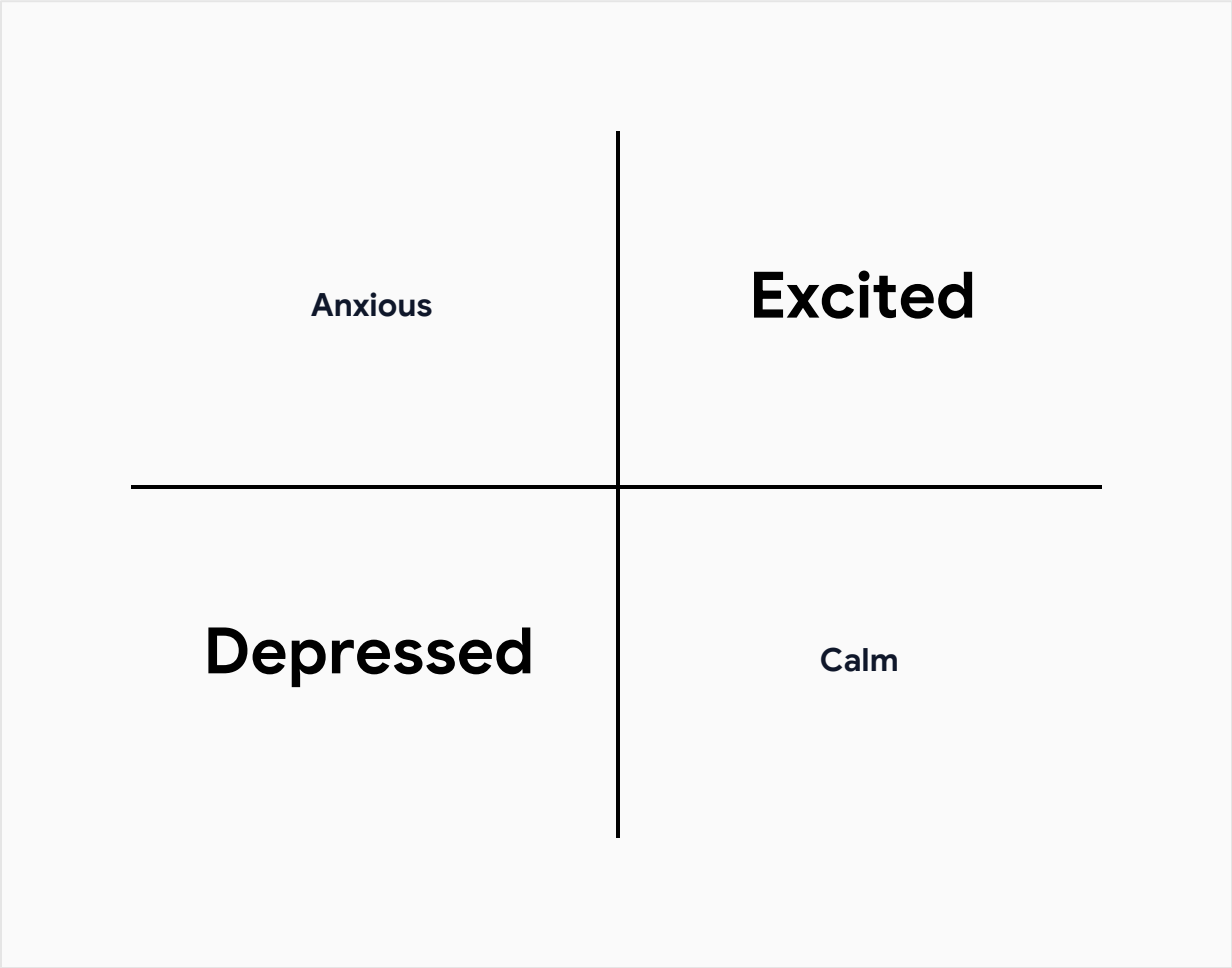
ID	Identified Risk	Impact	Technical Mitigation
R01	Cross-tenant data leakage	Critical	Enforce strict PostgreSQL Row Level Security (RLS). Ban ORM-level application filtering.
R02	Retail network offline state	High	Implement IndexedDB custom write buffers. Pre-fetch 4-hour audio queues locally.
R03	Copyright audit failure	High	Enforce mandatory AI-tagging and metadata validation prior to timeline assignment.
R04	AI Prompt Injection / Hallucination	Medium	Use strict JSON Object schemas (`response_format`) and temperature limiting on LLM.

Valence-Arousal Psychology

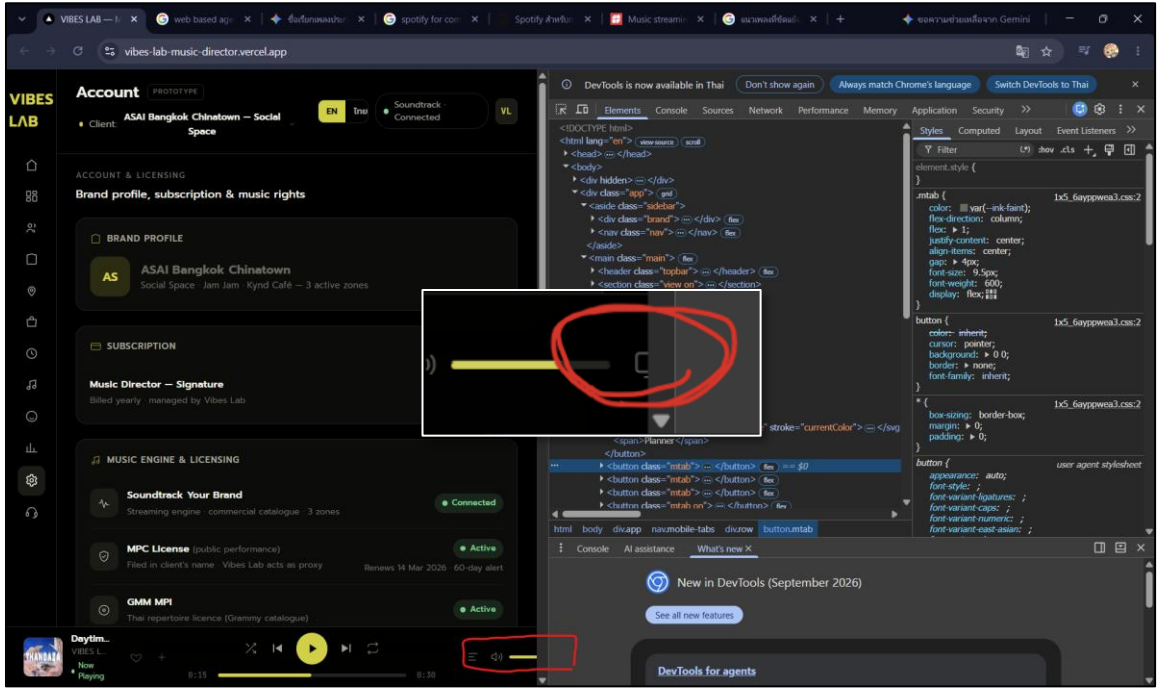
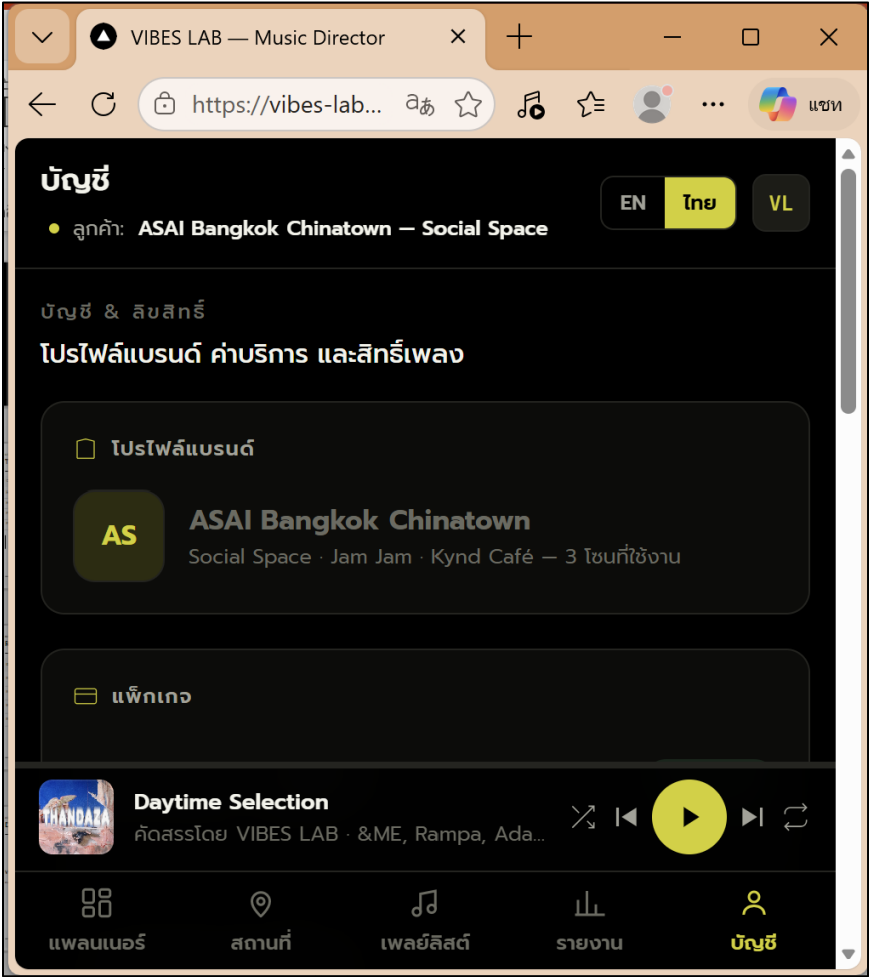
Audio Deep Learning: Extracts high-level features from frequency and tempo.

Valence (X-Axis): Measures emotional positivity (Sad to Happy).

Arousal (Y-Axis): Measures energy intensity (Calm to Energetic).



ISSUES ON THE WEBSITE



END OF PRESENTATION



VIBES LAB PARFUMS